BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: PR - XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 18

Note: [Urdu instructions printed at top of paper — best-effort] ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ جواب میں ایک سے زیادہ دائرے بھرے ہوں تو جواب غلط تصور ہوگا۔

Q1. Grignard's reagent contains _____ metal.

(A) Fe (B) Ca

(C) Mg (D) Mn

Q2. The bond angles of H-C-H and H-C-Cl bonds are _____.

(A) Less than 109.5° (B) Greater than 109.5°

(C) Equal to 109.5° (D) Equal to 120°

Q3. 1,2,4,5-tetramethylbenzene is the IUPAC name of _____.

(A) Xylene (B) Toluene

(C) Durene (D) Mesitylene

Q4. Opticla activity of a compound is measured by an instrument called _____.

(A) Photometer (B) Polarimeter

(C) Refractometer (D) Calorimeter

Q5. All of the given are alkanes except _____.

(A) C_3H_8 (B) C_4H_{10}

(C) C_5H_{12} (D) C_6H_6

Q6. Oxa is the functional group of _____.

(A) Ester (B) Aldehyde

(C) Ether (D) Ketone

Q7. Which of the given is not organic in nature?

(A) $NH_2CO\text{-}NH_2$ (B) $CH_3\text{-}OH$

(C) NH_4CNO (D) CH_3Cl

Q8. The most stable oxidation state of manganese in acidic or neutral solution is _____.

(A) +2 (B) +4

(C) +6 (D) +7

Q9. Al_2O_3 is _____.

(A) Acidic oxide (B) Neutral oxide

(C) Basic oxide (D) Amphoteric oxide

Q10. Which of the given is NMR inactive nucleus?

(A) ^{31}P (B) ^{2}H

(C) ^{15}N (D) ^{16}O

Q11. In combustion analysis CO_2 is absorbed by _____.

(A) $Mg(ClO_4)_2$ (B) KOH

(C) CuO (D) $CuSO_4$

Q12. The height of the stratosphere is from _____.

(A) 1 to 10 km (B) 50 to 85 km

(C) 10 to 50 km (D) 85 to 387 km

Q13. Total number of chlorine atoms in DDT are _____.

(A) 3 (B) 4

(C) 5 (D) 6

Q14. Proteins are polymers of _____.

(A) Glucose (B) Peptide bonds

(C) Disulphide bridges (D) Amino acids

Q15. All are the derivatives of carboxylic acid except _____.

(A) Acid Halides (B) Alkyl Halides

(C) Nitrile (D) Ester

Q16. Gem-Diol is the product of acid/base catalyzed reaction of water with _____.

(A) Aldehyde/ketone (B) Alcohol

(C) Carboxylic acid (D) Ether

Q17. In Williamson's synthesis _____ pair of reactants, react with each other to form ether.

(A) R-OH and Na
(B) R-OH and R-O Na^+
(C) R-OH and R-Mg-X
(D) R-OH and R-OH

Q18. Lucas reagent is _____.

(A) R-OH and HCl
(B) R-OH and $ZnCl_2$
(C) HCl and $ZnCl_2$
(D) None of these

SECTION-B

Time Allowed: 2 Hours and 40 minutes

QQ2. Answer any TEN parts. Each part carries FOUR marks.

(1) Write the chemical reactions to show that Al_2O_3 is amphoteric in nature?

(2) Define electron affinity. Discuss its variation in Halogens?

(3) Why compounds of Cu^+ are diamagnetic while Cu^{2+} are paramagnetic?

(4) What is homologous series? Give the characteristics of homologous series?

(5) Enlist four physical properties of cycloalkanes?

(6) The reaction of H-Br with H-C≡C-H gives 1,1-dibromoethane while with CH_3-C≡C-H, it gives 2,2-dibropropane. Give reason.

(7) Write the IUPAC names of the following. (a) $C(CH_3)_4$ (b) CH_3-$CH(CH_3)$-CH(OH)-CH_3 (c) $(CH_3)_3$C-CH_2-C(=O)-H (d) HOOC-CH_2-COOH

(8) Why secondary amines are more basic than tertiary amines in aqueous solution?

(9) Steric hindrance makes the ketones less reactive than aldehydes. Comment over it.

(10) Discuss the reactions of phenol with; (a) Bromine (b) Sodium metal

(11) What is the effect of substituents on the acidity of carboxylic acids, if attached with carboxylic acid molecule?

(12) State the medical problems that may relate to deficiencies of calcium and phosphorus?

(13) Give the applications of infrared spectroscopy.

SECTION-C

Note: Attempt any THREE questions. All questions carry equal marks.

QQ3.

(a) What is acid rain? Discuss its effects on environment and living organisms?

(b) Describe four fractions obtained by fractional distillation of petroleum?

QQ4.

(a) What are elimination reactions? Discuss E^1 mechanism in detail.

(b) How are aldehydes and ketones reduce to; (i) Alkanes by Wolf-Kishner Reduction (ii) Alcohols by Catalytic Method

QQ5.

(a) Define stereoisomerism and explain its types?

(b) How carbon and hydrogen can be detected in organic compounds?

QQ6.

(a) Draw the structural formula of the following compounds; (i) Naphthalene (ii) Acetic Anhydride (iii) Adipic Acid (iv) 2,4,6-Trinitrotoluene (v) 3,4-dimethylhexan-2-ol

(b) Differentiate between partial and total synthesis.